Lecture 3 — Implied Volatility

Study notes: definition, existence and uniqueness, root-finding (bisection, Newton), the volatility surface, and the deficiencies of Black–Scholes

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Abstract

These notes work through Lecture 3 in full. Implied volatility is the value of the single free Black–Scholes parameter σ that reproduces an observed market option price. The lecture has two halves. First, the *theory*: $\sigma \mapsto C^{\mathrm{BS}}(\sigma)$ is continuous, strictly increasing (because $\mathcal{V} > 0$) with explicit limits as $\sigma \to 0^+$ and $\sigma \to \infty$, so the inverse exists and is unique on the admissible price interval. Second, the *numerics*: bisection (robust, linear) and Newton (fragile but quadratic), with the key practical result that starting Newton at the inflection point $\hat{\sigma}$ of $C(\sigma)$ gives *global* monotone convergence. Finally, the empirical observation that implied volatility is not flat across strike and maturity — the smile/skew/surface — which is the diagnostic that constant-volatility Black–Scholes is misspecified, and the motivation for the local-, stochastic-volatility and jump models of later lectures. Everything is derived; where I fill gaps not on the slides (notably the closed form of $\mathcal{V}$, the second-derivative identity, and the put treatment) I say so explicitly.

Contents

1 Setup: the four knowns and the one unknown

We work inside the Black–Scholes model. Under the risk-neutral measure $\mathbb{Q}$ the stock solves $\mathrm{d}S_t = rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t^{\mathbb{Q}}$, and the time-$t$ price of a European call with strike K and maturity T is the closed form

$$C^{\mathrm{BS}}(S_t, t; \sigma) = S_t\,N(d_1) - Ke^{-r\tau}N(d_2), \qquad \tau := T - t, \tag{1}$$

with

$$d_1 = \frac{\ln(S_t/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = \frac{\ln(S_t/K) + (r - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}} = d_1 - \sigma\sqrt{\tau}, \tag{2}$$

and N the standard normal CDF, $N'(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$.

The formula has five inputs $(\tau, K, r, S_t, \sigma)$. Four of them are directly observable: the time to maturity $\tau = T - t$, the strike K, the risk-free rate r, and the current spot S_t. The fifth, the volatility σ, is *not* observable: it is a property of the unobserved future, not a quoted number. Black–Scholes treats it as a constant model parameter, but the market does not hand it to us.

The inverse problem. The market *does* hand us option prices C^{mkt}. So we turn (**??**) around. Holding (τ, K, r, S_t) fixed at their observed values, regard the call price as a scalar function of the one free parameter,

$$\sigma \;\longmapsto\; C^{\mathrm{BS}}(\sigma),$$

and ask: which σ reproduces the quote? The *implied volatility* σ^{impl} is defined as the solution of

$$V^{\mathrm{BS}}(\sigma^{\mathrm{impl}}; r, \tau, K, S_t) - V^{\mathrm{mkt}} = 0, \tag{3}$$

i.e. the volatility "implied" by the market price through the Black–Scholes lens. It is the market's quote expressed in the units of the model. We treat the call below; the put is identical after one remark (Section **??**).

Remark 1. *Implied volatility is* not *a forecast of realised volatility and* not *a directly measured quantity. It is a change of variable: a one-to-one re-quoting of a price as a vol, using a model we already suspect is wrong. Its usefulness is precisely that, were Black–Scholes true, σ^{impl} would be the same constant for every (K, T). The extent to which it is* not *constant is a direct readout of the model's misspecification.*

2 Existence and uniqueness of the implied volatility

For (**??**) to define σ^{impl} unambiguously we need: (i) $C^{\text{BS}}(\sigma)$ strictly monotone in σ (so the inverse is single-valued), and (ii) the market price to lie in the range of C^{BS}. Both follow from elementary properties of (**??**).

2.1 Vega is strictly positive

The *vega* is the sensitivity of the price to volatility,

$$\mathcal{V} := \frac{\partial C}{\partial \sigma}.$$

Differentiate (**??**). By the chain rule, treating d_1, d_2 as functions of σ,

$$\frac{\partial C}{\partial \sigma} = S_t N'(d_1)\,\frac{\partial d_1}{\partial \sigma} - Ke^{-r\tau} N'(d_2)\,\frac{\partial d_2}{\partial \sigma}. \tag{4}$$

From $d_2 = d_1 - \sigma\sqrt{\tau}$ we get $\dfrac{\partial d_2}{\partial \sigma} = \dfrac{\partial d_1}{\partial \sigma} - \sqrt{\tau}$. The key simplification is the identity

$$S_t\, N'(d_1) = Ke^{-r\tau}\, N'(d_2), \tag{5}$$

which I prove below. Substituting (**??**) into (**??**), the two terms carrying $\partial d_1/\partial\sigma$ have equal coefficients $S_t N'(d_1) = Ke^{-r\tau}N'(d_2)$ and cancel:

$$\frac{\partial C}{\partial \sigma} = S_t N'(d_1)\frac{\partial d_1}{\partial \sigma} - Ke^{-r\tau}N'(d_2)\Big(\frac{\partial d_1}{\partial \sigma} - \sqrt{\tau}\Big) = Ke^{-r\tau}N'(d_2)\,\sqrt{\tau}.$$

Using (**??**) once more to rewrite the prefactor in terms of S_t and d_1,

$$\boxed{\mathcal{V} = \frac{\partial C}{\partial \sigma} = S_t\, N'(d_1)\,\sqrt{\tau} = \frac{S_t\sqrt{\tau}}{\sqrt{2\pi}}\, e^{-\frac{1}{2}d_1^2} > 0} \qquad (\sigma > 0,\ \tau > 0). \tag{6}$$

Every factor on the right is strictly positive, so $\mathcal{V} > 0$. Hence $C^{\text{BS}}(\sigma)$ is *strictly increasing* in σ on $(0, \infty)$. This is the structural fact that makes the implied volatility well posed: a strictly monotone continuous function has a continuous inverse, so *at most one* σ^{impl} can solve (**??**).

Proof of identity (??). (Not on the slides; supplied here because Fang is exactly the kind of examiner who asks "why does that identity hold?") Write the ratio

$$\frac{N'(d_1)}{N'(d_2)} = \exp\Big(-\tfrac{1}{2}(d_1^2 - d_2^2)\Big) = \exp\Big(-\tfrac{1}{2}(d_1 - d_2)(d_1 + d_2)\Big).$$

Now $d_1 - d_2 = \sigma\sqrt{\tau}$ and $d_1 + d_2 = \dfrac{2\ln(S_t/K) + 2r\tau}{\sigma\sqrt{\tau}}$ (the $\pm\frac{1}{2}\sigma^2\tau$ terms cancel in the sum). Therefore

$$-\tfrac{1}{2}(d_1 - d_2)(d_1 + d_2) = -\tfrac{1}{2}\,\sigma\sqrt{\tau}\cdot\frac{2\ln(S_t/K) + 2r\tau}{\sigma\sqrt{\tau}} = -\ln(S_t/K) - r\tau,$$

so $\dfrac{N'(d_1)}{N'(d_2)} = e^{-\ln(S_t/K) - r\tau} = \dfrac{K}{S_t}\,e^{-r\tau}$, i.e. $S_t N'(d_1) = Ke^{-r\tau}N'(d_2)$. $\square$

Remark 2 (Vega's shape). $\mathcal{V} = S_t\sqrt{\tau}\,N'(d_1)$ *is a Gaussian bump in* d_1*, maximised when* $d_1 = 0$*, i.e. near the money. It decays to zero for deep in- or out-of-the-money options (*$|d_1| \to \infty$*). This will matter twice below: it is why Newton can stall in the wings (tiny denominator), and it is the reason the implied-vol* inversion *is ill-conditioned far from the money — a small price error is divided by a near-zero vega and blows up the implied vol.*

2.2 The admissible price interval

Monotonicity gives *at most* one solution. Existence requires the quote to lie in the range of C^{BS}, which we now pin down by taking the two limits in σ.

Lower limit, $\sigma \to 0^+$. As $\sigma \to 0^+$ the sign of d_1, d_2 is governed by the sign of $\ln(S_t/K) + r\tau$ (the volatility-independent part of the numerator), since $\sigma\sqrt{\tau} \to 0$ in the denominator amplifies that sign:

- If $\ln(S_t/K) + r\tau > 0$: $d_1, d_2 \to +\infty$, $N(d_1), N(d_2) \to 1$, so $C(\sigma) \to S_t - Ke^{-r\tau} > 0$.
- If $\ln(S_t/K) + r\tau < 0$: $d_1, d_2 \to -\infty$, $N(d_1), N(d_2) \to 0$, so $C(\sigma) \to 0$.
- If $\ln(S_t/K) + r\tau = 0$: $d_1, d_2 \to 0$, $N \to \frac{1}{2}$, so $C(\sigma) \to \frac{1}{2}(S_t - Ke^{-r\tau}) = 0$, the last equality because the case condition is exactly $S_t = Ke^{-r\tau}$.

The three branches collapse to one statement,

$$\lim_{\sigma \to 0^+} C(\sigma) = \max\bigl(S_t - Ke^{-r\tau},\, 0\bigr), \tag{7}$$

which is the no-arbitrage *intrinsic value* of the call (its forward-discounted moneyness, floored at zero). This is the natural floor: a zero-vol stock grows deterministically at r, and the call is worth its discounted in-the-money amount or nothing.

Upper limit, $\sigma \to \infty$. As $\sigma \to \infty$, the $+\frac{1}{2}\sigma^2\tau$ term sends $d_1 \to +\infty$ while the $-\frac{1}{2}\sigma^2\tau$ term sends $d_2 \to -\infty$. Hence $N(d_1) \to 1$, $N(d_2) \to 0$, and

$$\lim_{\sigma \to \infty} C(\sigma) = S_t. \tag{8}$$

Economically: with infinite volatility the downside is unchanged (payoff floored at 0) but the upside is unbounded, and the option's value rises to the whole spot S_t (the discounted strike becomes negligible against the dispersion). It is a supremum, not attained.

Conclusion. Combining (**??**), (**??**), (**??**): for every $\sigma \geq 0$,

$$\max\bigl(S_t - Ke^{-r\tau}, 0\bigr) \;\leq\; C(\sigma) \;<\; S_t, \tag{9}$$

and C is a continuous strictly increasing bijection from $[0, \infty)$ onto $\bigl[\max(S_t - Ke^{-r\tau}, 0),\, S_t\bigr)$. Therefore:

> *If* $C^{\mathrm{mkt}} \in \bigl[\max(S_t - Ke^{-r\tau}, 0),\, S_t\bigr)$, then there is a *unique* $\sigma^{\mathrm{impl}} \geq 0$ with $C(\sigma^{\mathrm{impl}}) = C^{\mathrm{mkt}}$.

A quote outside this band violates a static no-arbitrage bound (a call worth less than intrinsic, or more than the stock) and admits no real implied volatility — which in practice flags a stale or erroneous quote, or a quote at the bid/ask edge.

3 Numerical inversion

The equation $F(\sigma) := C(\sigma) - C^{\mathrm{mkt}} = 0$ has no closed-form solution (it sits behind the nonlinear $N(d_1), N(d_2)$), so we root-find numerically. Two methods are on the slides.

3.1 Bisection

Idea. If F is continuous and changes sign on $[\sigma_a, \sigma_b]$, i.e. $F(\sigma_a)F(\sigma_b) < 0$, the intermediate value theorem guarantees a root inside. Evaluate the midpoint $\sigma_m = \frac{1}{2}(\sigma_a+\sigma_b)$; the sign of $F(\sigma_m)$ agrees with exactly one endpoint, so one of $[\sigma_a, \sigma_m]$, $[\sigma_m, \sigma_b]$ still brackets the root. Replace the bracket by that half and repeat.

Pseudocode (as on the slide).

```
N <- 1
while N <= NMAX:
    c <- (a+b)/2                      # midpoint
    if F(c)=0 or (b-a)/2 < TOL:       # converged
        output c; stop
    N <- N+1
    if sign(F(c)) = sign(F(a)): a <- c    # root in [c,b]
    else:                       b <- c    # root in [a,c]
```

Why it always works here. By (**??**) and monotonicity, an initial bracket exists whenever the quote is admissible: take σ_a tiny (then $F < 0$ once $C^{\mathrm{mkt}} >$ intrinsic) and σ_b large (then $F > 0$ once $C^{\mathrm{mkt}} < S_t$). Monotonicity of C guarantees a single sign change, so bisection cannot be confused by spurious roots.

Convergence. The bracket halves each step: after n iterations the root is localised to width $(\sigma_b - \sigma_a)/2^n$. So the error obeys $|\sigma_n - \sigma^*| \le 2^{-n}(\sigma_b - \sigma_a)$ — *linear* convergence with rate 1/2, one bit of accuracy per iteration. To reach tolerance ε needs about $\log_2\big((\sigma_b - \sigma_a)/\varepsilon\big)$ steps.

Trade-off. Advantage: needs no derivative and no good initial guess — only a sign- changing bracket, which we can always produce. It is globally convergent and bullet-proof. Disadvantage: only linear, so it is slow when high accuracy is wanted.

3.2 Newton–Raphson

Derivation. To solve $F(\sigma^*) = 0$, Taylor-expand F about the current iterate σ_n for a small step δ:

$$F(\sigma_n + \delta) = F(\sigma_n) + \delta F'(\sigma_n) + \mathcal{O}(\delta^2).$$

Drop the quadratic remainder and demand the linear model vanish, $F(\sigma_n)+\delta F'(\sigma_n) = 0$. Solving for δ and setting $\sigma_{n+1} = \sigma_n + \delta$ gives the iteration

$$\sigma_{n+1} = \sigma_n - \frac{F(\sigma_n)}{F'(\sigma_n)} = \sigma_n - \frac{C(\sigma_n) - C^{\mathrm{mkt}}}{\mathcal{V}(\sigma_n)}. \tag{10}$$

Here $F' = C' = \mathcal{V}$ from (**??**), so each step needs one price and one vega evaluation — both cheap, and both already available in closed form. Geometrically, σ_{n+1} is where the tangent to F at σ_n crosses zero.

Local convergence theorem. If $F \in C^2$ and there is a simple root σ^* with $F(\sigma^*) = 0$, $F'(\sigma^*) \neq 0$, then there is $\delta > 0$ such that for any start with $|\sigma_0 - \sigma^*| < \delta$ the iterates are well defined, $\sigma_n \to \sigma^*$, and there is a constant C with

$$|\sigma_{n+1} - \sigma^*| \le C\,|\sigma_n - \sigma^*|^2. \tag{11}$$

This is *quadratic* convergence: the number of correct digits roughly doubles each step. For the IV problem $F' = \mathcal{V} > 0$ at any interior root, so the simple-root hypothesis holds.

Why quadratic. Subtract σ^* from (**??**) and use $F(\sigma^*) = 0$:

$$\begin{aligned}\sigma_{n+1} - \sigma^* &= \sigma_n - \sigma^* - \frac{F(\sigma_n)}{F'(\sigma_n)} = \sigma_n - \sigma^* - \frac{F(\sigma_n) - F(\sigma^*)}{F'(\sigma_n)} \\ &= \sigma_n - \sigma^* - \frac{(\sigma_n - \sigma^*)F'(\sigma_n) + \mathcal{O}\big((\sigma_n - \sigma^*)^2\big)}{F'(\sigma_n)} = \mathcal{O}\big((\sigma_n - \sigma^*)^2\big),\end{aligned}$$

where the third equality is the Taylor expansion $F(\sigma_n) - F(\sigma^*) = (\sigma_n - \sigma^*)F'(\sigma_n) - \frac{1}{2}(\sigma_n - \sigma^*)^2 F''(\xi) + \cdots$ rearranged so the linear part cancels $(\sigma_n - \sigma^*)$. The leading surviving term is quadratic, with constant $\approx \frac{1}{2}|F''/F'|$ near σ^*.

Trade-off. Advantage: much faster than bisection (quadratic vs. linear). Disadvantage: *local* — a poor σ_0 can overshoot, diverge, or step to negative volatility, and in the deep wings $\mathcal{V}(\sigma_n) \approx 0$ makes the update (**??**) explode. So Newton is fast where it works but is not robust by itself.

3.3 Globalising Newton: start at the inflection point $\hat{\sigma}$

The slides give a clean cure for Newton's fragility: a special starting point from which convergence is *global and monotone.* The construction rests on the convexity structure of $C(\sigma)$.

Second derivative. Differentiating $\mathcal{V} = S_t\sqrt{\tau}\,N'(d_1)$ in σ and simplifying (using $\partial d_1/\partial\sigma = -d_2/\sigma$, a standard reduction) gives the slide's formula

$$\frac{\partial^2 C}{\partial\sigma^2} = \frac{S_t\sqrt{\tau}}{\sqrt{2\pi}}\, e^{-\frac{1}{2}d_1^2} \cdot \frac{d_1 d_2}{\sigma}, \qquad \sigma > 0. \tag{12}$$

The prefactor is positive, so the sign of $\partial^2 C/\partial\sigma^2$ is the sign of $d_1 d_2$. Now

$$d_1 d_2 = \frac{\ln(S_t/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}} \cdot \frac{\ln(S_t/K) + (r - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}} = \frac{\big(\ln(S_t/K) + r\tau\big)^2 - \big(\frac{1}{2}\sigma^2\tau\big)^2}{\sigma^2\tau},$$

a difference of squares. This vanishes precisely when $\big(\ln(S_t/K) + r\tau\big)^2 = \frac{1}{4}\sigma^4\tau^2$, i.e. at

$$\sigma = \hat{\sigma} := \sqrt{\frac{2\,\big|\ln(S_t/K) + r\tau\big|}{\tau}}. \tag{13}$$

For $0 \le \sigma < \hat{\sigma}$ the squared moneyness term dominates, $d_1 d_2 > 0$, so $\partial^2 C/\partial\sigma^2 > 0$ and C is *convex.* For $\sigma > \hat{\sigma}$ we have $d_1 d_2 < 0$, so $\partial^2 C/\partial\sigma^2 < 0$ and C is *concave.* Thus $\hat{\sigma}$ is the unique inflection point of $C(\sigma)$, and — crucially — it is where $\mathcal{V} = C'$ attains its *maximum* (the slope is largest exactly where curvature changes sign).

Global monotone convergence from $\sigma_0 = \hat{\sigma}$. Suppose the true root satisfies $\sigma^* > \sigma_0 = \hat{\sigma}$ (the case $\sigma^* < \hat{\sigma}$ is symmetric). By the mean value theorem applied to F on $[\sigma_0, \sigma^*]$, there is $\xi_0 \in (\sigma_0, \sigma^*)$ with $F(\sigma_0) - F(\sigma^*) = (\sigma_0 - \sigma^*)F'(\xi_0)$, hence from (**??**)

$$\sigma_1 - \sigma^* = (\sigma_0 - \sigma^*) - \frac{(\sigma_0 - \sigma^*)F'(\xi_0)}{F'(\sigma_0)} \implies \frac{\sigma_1 - \sigma^*}{\sigma_0 - \sigma^*} = 1 - \frac{F'(\xi_0)}{F'(\sigma_0)}.$$

Because $\sigma_0 = \hat{\sigma}$ *maximises* $F' = \mathcal{V}$, we have $0 < F'(\xi_0) \le F'(\sigma_0)$, so the ratio lies in $(0, 1)$. Therefore $\sigma_1 - \sigma^*$ has the same sign as $\sigma_0 - \sigma^*$ but smaller magnitude: $\sigma_0 < \sigma_1 < \sigma^*$. At the next step σ_1 lies on the concave branch where $F' = \mathcal{V}$ is positive and *decreasing*, and the same MVT argument with $\sigma_1 < \xi_1 < \sigma^*$ gives the ratio again in $(0, 1)$, so $\sigma_1 < \sigma_2 < \sigma^*$. Inductively $\sigma_n < \sigma_{n+1} < \sigma^*$: the iterates form an increasing sequence bounded above by σ^*, hence converge, and the limit is the root. The error decreases monotonically to zero and, by (**??**), quadratically near the end.

Remark 3. *The starting point (??) is the trick that makes Newton both fast and safe for this specific problem: the maximal-slope property of $\hat{\sigma}$ is exactly what forces the contraction ratio $1 - F'(\xi)/F'(\sigma_0)$ into $(0, 1)$ on the first step, after which the iterates stay on the "good" (concave, monotone) side. This is a problem-specific globalisation, not a general Newton guarantee. In practice one still caps iterations and falls back to bisection if a step leaves the admissible interval.*

3.4 Method comparison

	Bisection	**Newton (from $\hat{\sigma}$)**
Order of convergence	linear, rate $\frac{1}{2}$	quadratic
Needs derivative?	no	yes ($\mathcal{V}$, closed form)
Needs good start?	no, only a bracket	yes; $\hat{\sigma}$ supplies it
Robustness	global, fail-safe	global here by the $\hat{\sigma}$ argument
Cost per step	one price eval	one price + one vega eval

A common production choice is a hybrid: a few bisection steps to localise, then switch to Newton for the final digits; or Newton from $\hat{\sigma}$ with a bisection safety net.

4 The put, and a sanity check via parity

The put is handled identically. By put–call parity $C - P = S_t - Ke^{-r\tau}$, which is σ-independent, the put price is $P(\sigma) = C(\sigma) - (S_t - Ke^{-r\tau})$. Differentiating,

$$\frac{\partial P}{\partial \sigma} = \frac{\partial C}{\partial \sigma} = \mathcal{V} = S_t\sqrt{\tau}\, N'(d_1) > 0,$$

so the put has the *same* positive vega and the same strict monotonicity; its implied volatility is unique on the analogous admissible band, and an arbitrage-free call and put at the same (K, T) return the *identical* implied volatility. This is why one may quote one IV per (K, T) regardless of call/put. The same bisection/Newton machinery applies verbatim with C replaced by P.

5 The implied volatility surface

If Black–Scholes held exactly, σ^{impl} recovered from (??) would be a single constant, identical across every strike K and maturity T — because σ is, in the model, one fixed number. The empirical fact is that it is not. Inverting market quotes produces a function

$$(K, T) \longmapsto \sigma^{\text{impl}}(K, T),$$

the *implied volatility surface.* Its non-flatness is the central diagnostic of this lecture.

Empirical example (FTSE 100, 22 Aug 2001). With $S_0 = 5420.3$, $r = 0.05$, $T = \frac{1}{3}$ yr, inverting the quoted call prices across strikes $5125, \ldots, 5825$ yields implied volatilities that *fall* monotonically with strike (from ≈ 0.190 at $K = 5125$ to ≈ 0.174 at $K = 5825$) instead of staying flat. Under Black–Scholes they should coincide; they do not, so the market does not obey the model's constant-vol assumption.

5.1 Shapes in the strike direction

Holding T fixed, $K \mapsto \sigma^{\text{impl}}(K, T)$ takes characteristic shapes:

- **Smile.** A convex U: implied vol is lowest near the money and higher for both deep ITM and deep OTM strikes. Reading it through the model, the market prices *both* tails richer than a lognormal — consistent with a return distribution with heavier tails (excess kurtosis) than the Gaussian Black–Scholes assumes.

- **Skew (smirk).** A monotone decreasing curve: implied vol is high for low strikes (downside puts) and decays for high strikes. This is the typical equity-index shape and encodes a left-skewed (negatively skewed) risk-neutral distribution: crash risk is priced richer, so out-of-the-money puts carry higher implied vol.
- **Hockey stick.** A skew on the left blending into a smile on the right — a combination of the two, with a rich left tail and a modestly rich right tail.

5.2 Term structure in the maturity direction

Holding K fixed (say at the money), $T \mapsto \sigma^{\text{impl}}(K,T)$ is the *term structure.* The slide contrasts two models for $\mathrm{d}S/S = \mu\,\mathrm{d}t + \sigma(t)\,\mathrm{d}W_t$:

- Constant volatility σ_*: the ATM implied term structure is flat in T.
- Time-dependent $\sigma(t)$: the implied vol at maturity T is the root-mean-square of the instantaneous vol, $\sigma^{\text{impl}}(T)^2 = \frac{1}{T}\int_0^T \sigma(t)^2\,\mathrm{d}t$, so a rising $\sigma(t)$ produces an upward-sloping implied term structure. (This is the cleanest way to see that the BS implied vol over $[0,T]$ aggregates the path of instantaneous variance into a single average.)

5.3 The full surface

Combining both directions gives $\sigma^{\text{impl}}(K,T)$ over a 2-D domain. The slides show the generic pattern: a pronounced *smile at short maturities* (the wings are most curved when there is little time for the central limit theorem to Gaussianise the return), flattening into a milder *skew at longer maturities.* A flat plane would mean Black–Scholes; the curvature and tilt of the actual surface is the market's correction to it, and any candidate model is judged by how well it can reproduce this surface (calibration).

6 Why constant-vol Black–Scholes fails

The surface is a symptom; the lecture closes by naming the broken assumptions and the families of models that repair them.

Assumptions that fail in reality.

- **Continuous trading.** Delta hedging is assumed continuous-time and frictionless; real rebalancing is discrete, leaving hedging error.
- **No transaction costs.** Real hedging incurs costs, which feed back into prices.
- **Gaussian returns.** BS log-returns are normal; empirical returns are heavy-tailed and negatively skewed — exactly what the smile/skew reports.
- **Constant volatility.** Realised volatility is not constant: it clusters (calm and turbulent regimes persist), so a single σ cannot fit prices across (K,T) simultaneously.

Repair families (introduced here, developed later).

- **Term-structure volatility:** $\mathrm{d}S/S = \mu\,\mathrm{d}t + \sigma(t)\,\mathrm{d}W_t$ — deterministic $\sigma(t)$ fits the maturity dimension but, being deterministic, still gives a flat smile in strike at each T.
- **Local volatility:** $\mathrm{d}S/S = \mu\,\mathrm{d}t + \sigma(S,t)\,\mathrm{d}W_t$ — volatility a deterministic function of spot and time, calibrated (Dupire) to reproduce the *entire* surface exactly, at the cost of unrealistic forward smile dynamics.
- **Stochastic volatility:** $\mathrm{d}S/S = \mu\,\mathrm{d}t + \sigma(W^v,t)\,\mathrm{d}W_t^S$ with volatility itself driven by a second Brownian motion — e.g. Heston (Lecture 7). A genuinely random variance generates smiles/skews endogenously and is the workhorse for the COS method later.
- **Jump models:** adding a jump component (Merton, Lecture 8) injects discontinuities that fatten the short-maturity tails, fixing the steep short-dated smile that diffusion alone cannot.

These are precisely the models the rest of the course prices, and the implied-vol surface is the target they must hit.

7 Distinguish: implied vs. local vs. stochastic volatility

A point Fang may probe, since the slide lists $\sigma(t), \sigma(S,t), \sigma(W^v,t)$ side by side.

- **Implied volatility** is a *model-inversion output*, not a model input: a single number per quote, defined only through the Black–Scholes formula. It is the market price re-expressed in vol units. It says nothing on its own about the true dynamics.
- **Local volatility** $\sigma(S,t)$ is a *model input*: a deterministic surface that, by Dupire, can be chosen to match all implied vols at once. It is the "effective" instantaneous vol consistent with today's surface.
- **Stochastic volatility** $\sigma(W^v,t)$ is also a model input but *random*, with its own driving noise (and, in Heston, correlation ρ with the spot's noise). It explains *why* the surface has the shape and dynamics it does, rather than merely fitting it.

In one line: implied vol describes the data, local vol fits it deterministically, stochastic vol explains it.

Missing / not on the slides (flagged honestly)

- The closed form $\mathcal{V} = S_t\sqrt{\tau}\,N'(d_1)$ (**??**) is *derived* here; the slides assert $\mathcal{V} > 0$ and the cancellation but stop short of the boxed formula. The proof of the identity (**??**) is mine; the slide only states it "follows from $d_2 = d_1 - \sigma\sqrt{\tau}$ and the exponential form of N'".
- The reduction $\partial d_1/\partial\sigma = -d_2/\sigma$ used to get (**??**) is stated without derivation on the slides; I quote it.
- The put treatment via parity (Section **??**) is my addition; the slides only say "an analogous treatment applies".
- The RMS reading of the term structure, $\sigma^{\text{impl}}(T)^2 = \frac{1}{T}\int_0^T \sigma(t)^2\,\mathrm{d}t$, is the standard interpretation of the $\sigma(t)$ figure but is not written on the slide.
- The Dupire/local-vol and Heston cross-references are forward pointers to Lectures 7–8, not content of Lecture 3.